

READ FIRST — PSC program state (2026-09-08 remediation)

This file supersedes `SESSION_2026_06_10_SUMMARY.md`
as the read-first document.

That summary predates the June 13–15 work and the
six-round external review; reading it alone
reconstructs a v13-era state that asserts a broken
theorem as proved. Read this, then

`SESSION_2026_06_13_LEDGER.md`, then

`PROJECT_AUDIT_2026_09_08.md`.

Canonical manuscripts

Document	Status
PSC_PROOF_v15.tex/.pdf (20 pp)	CANONICAL. Six external review rounds; scope-clean. Architecture: B_σ finite (G1) + SCC Producer $\Rightarrow$ PDS.
Geometric_Mass_Balance_Two_Anchor_v9 (15 pp)	Companion (cohomological)

Document	Status
	/ transversal-substitution line). $v_9 = v_8 +$ one bridge-converse cross-reference.
PSC_PROOF_v14	SUPERSEDED with known error: states finiteness (thm:finiteness) as a proved theorem via the withdrawn predecessor contraction. Do not cite.
PSC_PROOF_v13 and earlier	SUPERSEDED with the same error. Do not cite.
"PSC_PROOF_v16 (29 pp)"	Does not exist. A memory entry referred to it; no file has ever been

Document	Status
	found. Treat as unreliable.

What is proved (v15 labels – every result carries exactly one)

- Theorem-grade: defect theorem $\Rightarrow$ UD; UD for powers; local witness injectivity; mass-balance K_2 obstruction; closed-nonproductive algebraic embedding $M_\sigma \subseteq N_C$; alphabet-3 wedge bound $\rho(N_C) \geq \beta|\beta_2|$ for **closed** wedge-nonzero SCCs.
- Hypothesis G1: finiteness of B_σ (unconditional only for $|A|=2$ and verified families).
v5 Theorem 5.1 is withdrawn – $\beta \cdot L(s') \leq L(s) + D$ is false (ratio 8.0, unbounded).
- Observation/diagnostic: phase-automaton padding (only legally-repeatable complete-cutting cycles are excluded; nonzero-offset recurrence uncontrolled – 19% empirically).
- Conjecture: SCC Producer (conj :producer) – the main open target.
- Conditional (companion): existence of a qualifying wedge-nonzero SCC (β -richness / pure- β -zero).
- Main result: **boxed conditional** B_σ finite + SCC Producer $\Rightarrow$ PDS.

The open core (unchanged since June)

Symbolic side: Conjecture conj:producer (no closed nonproductive SCC). Cohomology side: Q-A condition (1)'s upper bound $r \leq C_{\text{comp}} - 1$ (homological-Pisot rigidity; non-generic; no census purchase). Both sit behind the no-coarse-quotient meta-theorem. G1 open for $|A| \geq 3$.

Certificate / census figures (verified against files)

- H3' per-specimen certificates: **1,083 certified / 50 capped / 3 timeouts** (latest ledger in H3PRIME_LFIX_REDUCTION.md). A memory figure of "1,266" is unsupported by any file.
- Exhaustive $k=3$ PIP census: **4,554** specimens (BPA termination 4,554/4,554, 0 caps; BD-complex b_1 census 4,554 rows). This is a census SIZE, not a certificate count.
- Spectral band $\rho(N_C) \in [\beta|\beta_2|, \beta)$ on recurrent noncoincident SCCs; no closed nonproductive SCC, no trapped pair, no trapped triple in the corpus (elimination, not validation).

Instrument corrections you MUST use (June 13; only in this bundle until uploaded)

- `legal_swaps.py` — G2/structural BPA runs must seed from legality-filtered swaps; all-swap seeding over-counts recurrent SCCs in ~2% of specimens. Termination is seed-invariant.
- `exact_lengths.py` — exact- $\mathbb{Q}(\beta)$ genuineness; float census verified sound (0 mismatches).
- Offset-propagation matrix is $\mathbf{M}$ ($M[i,j] = \#i \text{ in } \sigma(j)$), **not** $\mathbf{M}^T$ — see `trapped_scc_search.py`.

Do-not-re-attempt (all refuted or dead-ended; details in the ledger)

H3 letter-graph / affine-fixed-point trapped detector
(transpose bug + wrong object) ·
Collar Certification Lemma (refuted; $R(C)$ tracks cycle
maxlen, not L_σ) · Q-B spectral
over-constraint (none; coupling is a bounded quotient) ·
condition (1) via census or $\times\beta$ lever ·
Barge–Olimb for 1D (no content) · kernel-graph ·
Rauzy/DT address · bilateral-permutative route ·
Rauzy tiling-boundary and radius-ceiling as exclusion
mechanisms.

Certificate documents — PATCHED

2026-09-08

PROOF_CERTIFICATE.md , V34_CLOSURE.md ,
DOMINANT_K2_SOURCE_V34.md in this bundle are the
corrected versions: "unconditional" → "conditional on
finite $B_\sigma(G1)$ "; v5 Thm 5.1 marked
withdrawn. The versions in the project knowledge (pre-
patch) carried a false "unconditional"
claim from June to September. Replace them.

Next work (per the sixth reviewer)

Mathematical, not editorial: the companion β -
richness/cyclic-vector route, or a direct attack on
SCC Producer / no diagonal-free zero-return systems.
Do not open another manuscript-scope pass.